

System Requirements Specification

High Priority Car Rental Management System (HPCRMS)

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Course Project — Software Engineering

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1. Introduction

1.1 Purpose

This purpose of this document is to define the functional and non-functional requirements for the High Priority Car Rental Management System (HPCRMS). The system aims to reduce customer waiting time by digitizing the reservation, rental, payment, pickup, and return processes.

1.2 Scope

The HPCRMS provides customers with the ability to:

- Register online
- Search available vehicles
- Reserve vehicles
- Complete digital check-in
- Make online payments
- Pick up vehicles quickly
- Return vehicles

The system also enables administrators and rental staff to manage customers, vehicles, rentals, maintenance, and reporting.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers, project managers, instructors, and stakeholders. Readers should review Sections 1 and 2 first, then Section 3 for detailed functional requirements and the appendices for analysis models and glossary terms.

1.4 References

- HPCRMS Vision Document
- Lesson 4 & 5 course materials — Use-Case Modeling and Requirements Analysis.
- Wiegers, K., Software Requirements Specification template (used as the structural basis for this document).

2. Overall Description

2.1 Product Perspective

HPCRMS serves as the central platform for all rental operations, integrating customer management, reservation management, fleet management, payment processing, reporting, and maintenance scheduling into one unified system. Branches connect to a centralized cloud database that keeps vehicle availability synchronized in real time. Secure API endpoints expose vehicle availability and payment processing to authorized third-party applications.

2.2 Product Features

Reservation Management

- Vehicle Search
- Vehicle Reservation
- Reservation Modification

Customer Management

- Registration
- Profile Management

Rental Processing

- Digital Check-in
- Vehicle Assignment
- Rental Agreement

Vehicle Management

- Vehicle Inventory
- Vehicle Availability

Payment Processing

- Online Payments
- Refund Processing

Return Processing

- Vehicle Return
- Damage Assessment

Fleet Maintenance

- Maintenance Scheduling

Reporting

- Revenue Reports
- Utilization Reports

2.3 User Classes and Characteristics

User Class	Characteristics
Customer	General public; renters. Uses the web or mobile app for reservation, check-in, payment, pickup. No technical expertise assumed;
Rental Agent	Branch staff. Frequent daily use; assists customers, performs manual identity verification and vehicle assignment overrides.
Branch Manager	Branch-level supervisory staff. Moderate frequency of use; reviews report and resolves escalated disputes.
Fleet Manager	Operations staff responsible for the vehicle fleet. Frequent use of inventory, assignment, conducts vehicle inspections, and maintenance functions.
Finance Department	Back-office staff. Periodic use for payment reconciliation and financial reporting.

User Class	Characteristics
System Administrator	IT/operations staff with elevated privileges. Infrequent use, limited to account and access management.

2.4 Operating Environment

Client Side

- Web Browser
- Android Application
- iOS Application

Server Side

- Spring Boot
- Java 21
- REST API

Database

- MySQL 8/PostgreSql

Hosting

- Cloud Platform

2.5 Design and Implementation Constraints

- All customer and payment data must be encrypted at rest and in transit.
- The system must support at least 1,000 concurrent users.
- Vehicle availability must be synchronized in real time across all branches.
- RESTful API architecture
- MySQL database required
- Role-based access control (RBAC)
- HTTPS encryption required
- Integration with third-party payment gateway

2.6 User Documentation

- User Manual.
- Administrator Guide.
- API Documentation.
- Maintenance Guide.
- Training Materials for branch staff.

2.7 Assumptions and Dependencies

- Internet connectivity is available at all rental branches.
- Customers possess valid driver's licenses.
- Vehicles are equipped with GPS tracking.

- Online payment gateways and SMS/email services remain available.
- Vehicle availability is updated in real time, and branches share a centralized database.

3. System Features

This section organizes the functional requirements of HPCRMS by system feature. Each feature corresponds to one of the major use cases identified in the use-case model (Appendix B), which provides the detailed actor interactions, flows, preconditions, and postconditions underlying the requirements stated here.

3.1 Manage Customer Account

3.1.1 Description and Priority

Allows a customer to register an account, log in, and update their profile so they can reserve, verify, check in, pay for, and manage vehicle rentals. Precondition: the customer has access to the HPCRMS web or mobile application and an internet connection.

Priority: High

3.1.2 Stimulus/Response Sequences

- Customer requests to register → System displays the registration form.
- Customer submits registration details → System validates uniqueness of email, creates the profile, and confirms success, or reports a duplicate-account or validation error.
- Customer submits login credentials → System authenticates and displays the customer dashboard, or reports invalid credentials.
- Customer requests to edit their profile → System displays an editable profile form.
- Customer submits profile changes → System validates and saves the updated profile.

3.1.3 Functional Requirements

REQ-1.1: The system shall provide a registration form capturing first name, last name, email, phone number, driver's license number (optional), and password.

REQ-1.2: The system shall reject registration if an account with the same email address already exists and shall display a duplicate-account message.

REQ-1.3: The system shall enforce a minimum password complexity policy at registration and on password changes.

REQ-1.4: The system shall authenticate a customer's email and password before granting access to the customer dashboard.

REQ-1.5: The system shall temporarily lock a customer account after 5 consecutive failed login attempts.

REQ-1.6: The system shall allow a logged-in customer to update all profile fields except the email address.

REQ-1.7: The system shall encrypt customer personal data at rest and in transit.

3.2 Search and Reserve Vehicle

3.2.1 Description and Priority

Allows a customer to search real-time vehicle availability across branches and create, modify, or cancel a reservation. Precondition: the customer is logged in.

Priority: High

3.2.2 Stimulus/Response Sequences

- Customer submits search criteria (branch, dates, category) → System displays available vehicles with specifications, pricing, and images within 2 seconds, or suggests alternatives if none match.
- Customer selects a vehicle and confirms → System calculates the estimated cost, reserves the vehicle, updates availability across branches, and returns a confirmation within 5 seconds.
- Customer requests to modify a reservation → System updates the reservation and recalculates cost.
- Customer requests to cancel a reservation → System releases the vehicle and applies the cancellation policy.

3.2.3 Functional Requirements

REQ-2.1: The system shall display real-time vehicle availability across all branches within 2 seconds of a search request.

REQ-2.2: The system shall suggest alternative dates, branches, or categories when no vehicle matches the search criteria.

REQ-2.3: The system shall calculate the estimated total cost of a reservation prior to confirmation.

REQ-2.4: The system shall generate a reservation confirmation within 5 seconds of a customer confirming a reservation.

REQ-2.5: The system shall synchronize vehicle inventory availability across all branches in real time.

REQ-2.6: The system shall allow a customer to modify the dates, category, or branch of an existing reservation.

REQ-2.7: The system shall allow a customer to cancel an existing reservation and apply the applicable cancellation policy.

REQ-2.8: The system shall support at least 1,000 concurrent users performing searches and reservations.

REQ-2.9: The system shall trigger a confirmation notification upon successful reservation, modification, or cancellation.

3.3 Verify Customer Identity

3.3.1 Description and Priority

Automates validation of a customer's driver's license and identity as part of digital check-in, replacing manual counter verification. Precondition: the customer has an active, confirmed reservation.

Priority: High

3.3.2 Stimulus/Response Sequences

- Customer uploads license photos and a selfie → System submits the data to the external Driver's License Verification Service and receives a validation result.
- Verification result is inconclusive, or the verification service is unavailable → Rental Agent manually reviews the documents and overrides the verification status.

3.3.3 Functional Requirements

REQ-3.1: The system shall accept an uploaded photo of the front and back of a customer's driver's license and a selfie image.

REQ-3.2: The system shall transmit license and selfie data to the external Driver's License Verification Service through a secure API.

REQ-3.3: The system shall reject check-in and notify the customer if the license is expired or invalid.

REQ-3.4: The system shall allow a Rental Agent to manually override an inconclusive or unavailable automated verification result.

REQ-3.5: The system shall record the identity verification status associated with each reservation.

3.4 Complete Digital Check-In

3.4.1 Description and Priority

Enables a customer to complete rental paperwork digitally, including reviewing and signing the rental agreement, prior to vehicle pickup. Precondition: a confirmed reservation exists and identity has been verified.

Priority: High

3.4.2 Stimulus/Response Sequences

- Customer initiates check-in → System confirms identity verification, then auto-generates the rental agreement.
- Customer reviews terms and applies an electronic signature → System stores the signed contract and issues a QR code/pickup code.
- Customer declines the contract terms → System cancels check-in and reverts the reservation to "Pending."

3.4.3 Functional Requirements

REQ-4.1: The system shall require successful identity verification before generating a rental agreement.

REQ-4.2: The system shall auto-generate a rental agreement using reservation and vehicle details.

REQ-4.3: The system shall capture an electronic signature from the customer on the rental agreement.

REQ-4.4: The system shall store the signed digital contract associated with the reservation.

REQ-4.5: The system shall generate a unique QR code/pickup code upon successful check-in.

REQ-4.6: The system shall allow a Rental Agent to complete check-in on behalf of a customer without application access.

REQ-4.7: The system shall revert the reservation status to "Pending" if the customer declines the contract terms.

3.5 Process Payment

3.5.1 Description and Priority

Processes rental charges, deposits, and refunds securely through an integrated third-party payment gateway and produces a digital receipt. Precondition: check-in is complete, or a vehicle is being returned and final billing is required.

Priority: High

3.5.2 Stimulus/Response Sequences

- Customer selects a payment method → System calculates total charges and forwards the transaction to the external Payment Gateway.
- Payment Gateway authorizes the transaction → System generates and emails a digital receipt.
- Payment is declined → System notifies the customer and allows selection of another payment method.

- A cancellation or return adjustment requires a refund → System issues a refund request to the Payment Gateway.

3.5.3 Functional Requirements

REQ-5.1: The system shall calculate total charges, including rental fee, taxes, add-ons, and deposit, prior to payment.

REQ-5.2: The system shall transmit payment transactions to the external Payment Gateway over an encrypted connection.

REQ-5.3: The system shall generate and email a digital receipt upon successful payment.

REQ-5.4: The system shall notify the customer and allow selection of an alternate payment method if a payment is declined.

REQ-5.5: The system shall automatically retry or flag for manual reconciliation any transaction where the Payment Gateway times out.

REQ-5.6: The system shall support issuing refunds to the original payment method through the Payment Gateway.

REQ-5.7: The system shall encrypt and handle all payment transactions consistent with payment card data-protection practices.

3.6 Pick Up Vehicle (QR Code Scan)

3.6.1 Description and Priority

Allows a checked-in customer to retrieve their assigned vehicle quickly by scanning a QR code, minimizing counter wait time. Precondition: digital check-in and payment are complete.

Priority: High

3.6.2 Stimulus/Response Sequences

- Customer scans the QR code/pickup code → System verifies check-in and payment status and assigns a vehicle if one is not already assigned.
- System confirms a vehicle is available → System displays parking location and access instructions and releases the vehicle.
- QR code is invalid or expired → System directs the customer to a Rental Agent for manual assistance.

3.6.3 Functional Requirements

REQ-6.1: The system shall validate a scanned QR code/pickup code against an active, checked-in, and paid reservation.

REQ-6.2: The system shall automatically assign the best available vehicle to a reservation if none has been assigned prior to pickup.

REQ-6.3: The system shall display the assigned vehicle's parking location and access instructions upon successful validation.

REQ-6.4: The system shall update the reservation status to "Active Rental" and the vehicle status to "Rented" upon vehicle release.

REQ-6.5: The system shall direct the customer to a Rental Agent for manual assistance if the QR code is invalid or expired.

3.7 Return Vehicle and Record Inspection

3.7.1 Description and Priority

Allows a customer to return a rented vehicle and enables a Rental Agent to record a digitized inspection at return, closing out the rental. Precondition: the vehicle is on an active rental.

Priority: low

3.7.2 Stimulus/Response Sequences

- Customer returns the vehicle and notifies staff → System displays the digital inspection form pre-populated with the pickup inspection.
- Rental Agent records condition, mileage, fuel level, and damage photos → System calculates additional charges and processes final payment or refund.
- Customer disputes the reported condition → System flags the case for Branch Manager review.

3.7.3 Functional Requirements

REQ-7.1: The system shall provide a digital vehicle-inspection form pre-populated with the pickup inspection data.

REQ-7.2: The system shall allow a Rental Agent to record vehicle condition, mileage, fuel level, and damage photographs at return.

REQ-7.3: The system shall compare the return inspection against the pickup inspection to calculate late, damage, and fuel charges.

REQ-7.4: The system shall process a final payment or refund adjustment based on the return inspection.

REQ-7.5: The system shall update vehicle status to “Available” or “Needs Maintenance” based on the inspection outcome, and close the reservation.

REQ-7.6: The system shall flag a reservation for Branch Manager review when a customer disputes the reported vehicle condition.

REQ-7.7: The system shall support a self-service return checklist submitted through the customer application prior to formal agent inspection.

3.8 Manage Vehicle Inventory and Assignment

3.8.1 Description and Priority

Enables staff to view, add, and update vehicles across the fleet, and enables the system to automatically assign the best available vehicle to a reservation. Precondition: the staff member is authenticated with an appropriate role.

Priority: High

3.8.2 Stimulus/Response Sequences

- Staff adds or updates a vehicle record → System validates and saves the record and updates availability across branches.
- A reservation reaches check-in or pickup → System automatically assigns the best available matching vehicle.
- The automatically assigned vehicle becomes unavailable → Rental Agent manually reassigns a different vehicle.

3.8.3 Functional Requirements

REQ-8.1: The system shall allow authorized staff to add a new vehicle record with category, branch, and status.

REQ-8.2: The system shall allow authorized staff to update an existing vehicle’s category, branch, or status.

REQ-8.3: The system shall automatically assign the best available vehicle to a reservation based on category, branch, and availability.

REQ-8.4: The system shall allow a Rental Agent to manually reassign a vehicle if the automatically assigned vehicle becomes unavailable.

REQ-8.5: The system shall synchronize vehicle inventory and status across all branches in real time.

3.9 Schedule Vehicle Maintenance

3.9.1 Description and Priority

Allows the Fleet Manager to schedule and track preventive and corrective maintenance for vehicles and to monitor maintenance status fleet-wide. Precondition: the Fleet Manager is authenticated with appropriate role permissions.

Priority: Medium

3.9.2 Stimulus/Response Sequences

- Fleet Manager schedules a maintenance appointment → System marks the vehicle unavailable for the maintenance window.
- Fleet Manager records completion of service → System returns the vehicle to “Available” status.
- A vehicle requires urgent maintenance mid-rental → System flags the reservation for Branch Manager review upon return.

3.9.3 Functional Requirements

REQ-9.1: The system shall allow a Fleet Manager to schedule a maintenance appointment for a vehicle.

REQ-9.2: The system shall mark a vehicle “Unavailable” and remove it from bookable inventory for the duration of a scheduled maintenance window.

REQ-9.3: The system shall return a vehicle to “Available” status upon recorded completion of maintenance.

REQ-9.4: The system shall flag a reservation for Branch Manager review when a vehicle requires urgent maintenance mid-rental.

REQ-9.5: The system shall generate maintenance alerts based on mileage or time-based thresholds.

3.10 Generate Reports and Analytics

3.10.1 Description and Priority

Provides dashboards and reports on rental trends, fleet utilization, revenue, and branch performance, including demand forecasting. Precondition: the manager is authenticated with appropriate role permissions.

Priority: Medium

3.10.2 Stimulus/Response Sequences

- Manager selects a report type and date range → System aggregates data across branches and displays the report/dashboard.
- Manager requests to export a report → System exports the report as PDF or Excel.

3.10.3 Functional Requirements

REQ-10.1: The system shall generate utilization, revenue, branch-performance, and demand-forecast reports for a manager-specified date range.

REQ-10.2: The system shall aggregate data from reservations, payments, fleet records, and vehicle-location data across all branches.

REQ-10.3: The system shall allow a manager to export a generated report as PDF or Excel.

REQ-10.4: The system shall incorporate real-time vehicle location data (Feature 3.13) into fleet-utilization reporting.

3.11 Manage Users and Access

3.11.1 Description and Priority

Allows the System Administrator to create, update, and deactivate staff user accounts and to assign role-based permissions across the system. Precondition: the administrator is authenticated with administrative privileges.

Priority: Medium

3.11.2 Stimulus/Response Sequences

- Administrator creates a staff account with an assigned role → System validates uniqueness, saves the account, and applies role-based access control.
- Administrator requests to deactivate an account → System revokes access after confirmation.

3.11.3 Functional Requirements

REQ-11.1: The system shall allow a System Administrator to create a staff account with an assigned role (Rental Agent, Branch Manager, Fleet Manager, or Finance Department).

REQ-11.2: The system shall reject account creation if the email/username is already in use and shall report a duplicate-account message.

REQ-11.3: The system shall enforce role-based access control restricting functionality according to the assigned role.

REQ-11.4: The system shall require multi-factor authentication for all employee accounts.

REQ-11.5: The system shall allow a System Administrator to deactivate an existing staff account and immediately revoke its access.

3.12 Receive Notifications (Email/SMS)

3.12.1 Description and Priority

Automatically sends confirmations, reminders, and status updates to customers via email and SMS whenever a reservation, check-in, payment, or pickup event occurs. Precondition: a triggering event has occurred.

Priority: Medium

3.12.2 Stimulus/Response Sequences

- A reservation, check-in, payment, or pickup event occurs → System composes and sends a notification via the external Notification Service.

3.12.3 Functional Requirements

REQ-12.1: The system shall send a confirmation notification via email and/or SMS when a reservation is created, modified, or cancelled.

REQ-12.2: The system shall send a notification when digital check-in and payment are completed.

REQ-12.3: The system shall send a receipt notification upon successful payment processing.

3.13 Track Vehicle Location

3.13.1 Description and Priority

Monitors real-time vehicle GPS location for fleet oversight and utilization analysis. Precondition: the vehicle is equipped with GPS tracking.

Priority: Medium

3.13.2 Stimulus/Response Sequences

- Fleet Manager or the reporting feature requests vehicle location data → System retrieves current location from the external GPS Vehicle Tracking System.

3.13.3 Functional Requirements

REQ-13.1: The system shall retrieve real-time vehicle GPS location data from the external GPS Vehicle Tracking System.

REQ-13.2: The system shall make current vehicle location data available to Fleet Managers and to the reporting/analytics feature.

4. Use-Case Model

4.1 Actors

The following actors interact with HPCRMS. Primary actors initiate use cases to achieve a goal; secondary actors are external systems that HPCRMS depends on to complete a use case.

Actor	Type	Description
Customer	Primary	An individual who registers, reserves, verifies identity, checks in, pays, picks up, and returns a rental vehicle.
Rental Agent	Primary	Branch staff who assist customers, perform manual verification/assignment when needed, and conduct vehicle inspections.
Branch Manager	Primary	Monitors branch operations, reviews reports, and resolves escalated disputes (e.g., inspection discrepancies).
Fleet Manager	Primary	Manages vehicle inventory, assignment rules, and maintenance scheduling; monitors fleet utilization.
Finance Department	Primary	Processes and reconciles payments and refunds; reviews financial reports.
System Administrator	Primary	Creates and manages staff user accounts and role-based access permissions.
Payment Service	Secondary / External	Third-party service that authorizes and processes online payments and refunds.
Driver's License Verification Service	Secondary / External	Third-party service that validates license authenticity and matches identity.
Notification Service	Secondary / External	Third-party email/SMS service used to deliver confirmations, reminders, and updates.

Actor	Type	Description
Vehicle Tracking Service	Secondary / External	Provides real-time vehicle location data used for fleet tracking and utilization reporting.

4.2 Use-Case Diagrams

To keep the diagrams readable, the use-case model is presented as two related diagrams: one covering the customer-facing rental, and one covering employees functions.

Figure 1 — Use-Case Diagram: Customer Rental

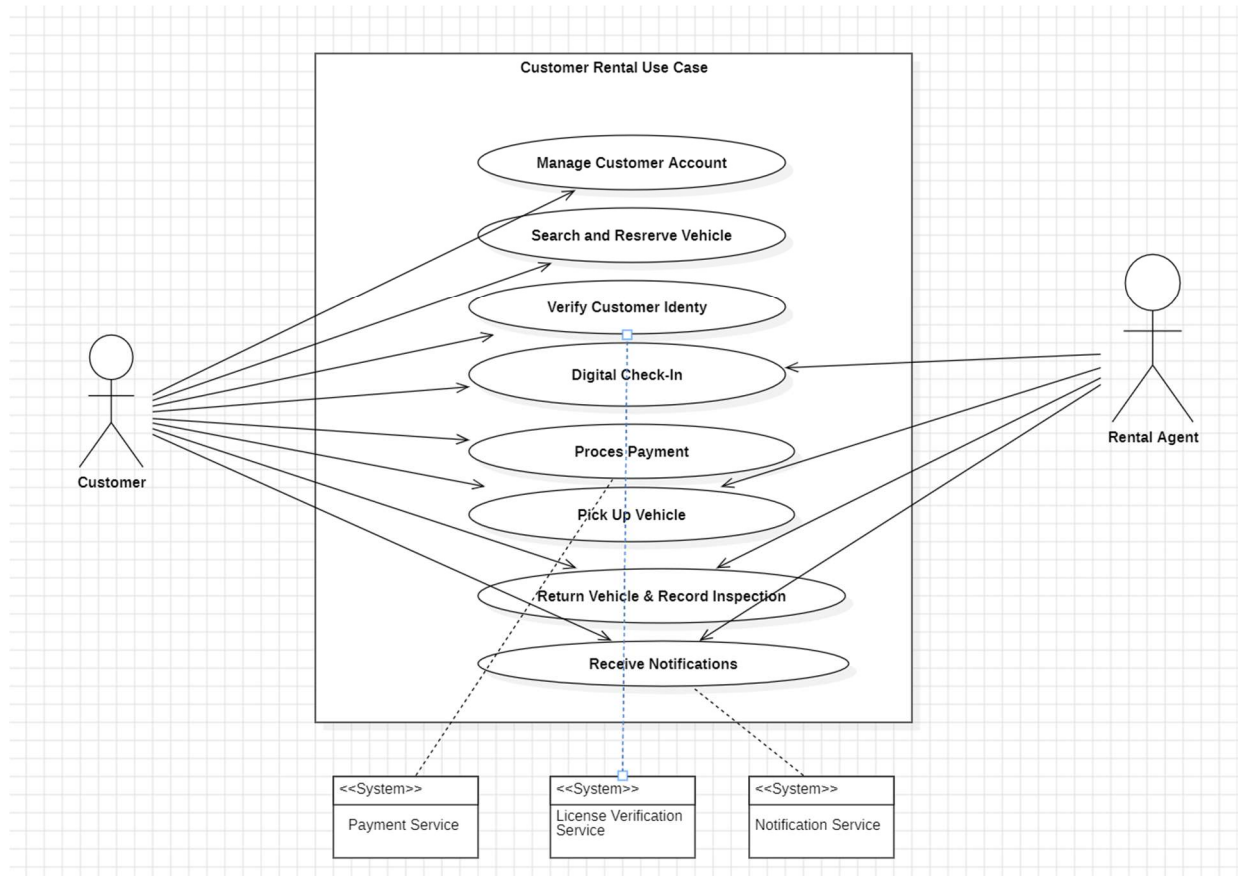


Figure 1. Customer registration through reservation, identity verification, digital check-in, payment, pickup, return, and notifications.

Figure 2 — Use-Case Diagram: Employee

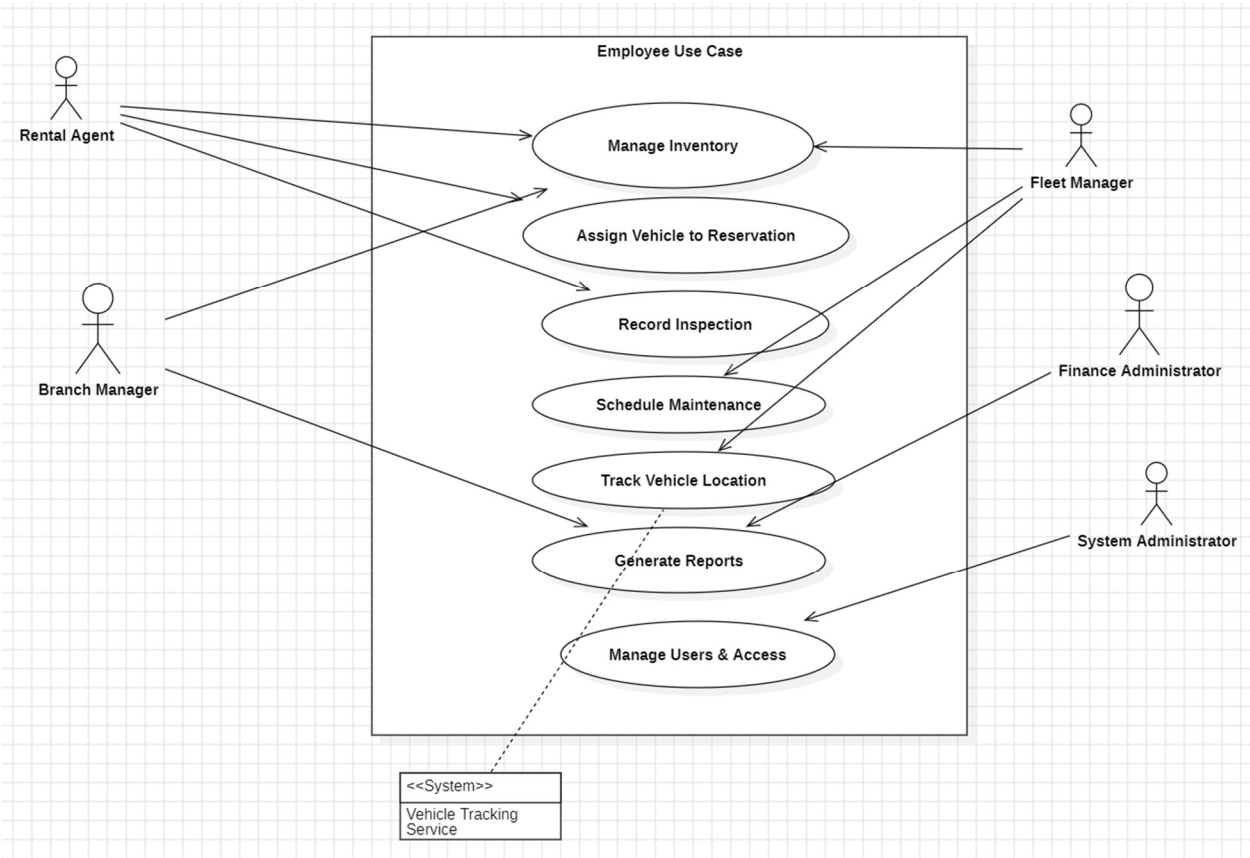


Figure 2. Staff-facing vehicle inventory, maintenance, reporting/analytics, and user-access administration.

4.3 Use-Case Descriptions

The major use cases identified from the problem statement and Vision Document.

UC-1: Manage Customer Account

Use Case Number: UC-1	
Name	Manage Customer Account
Description	Allows a customer to register a new account, log in, update profile information, and view rental history through the web or mobile application.
Actors	Customer
Preconditions	
Customer has access to the HPCRMS web or mobile application and an internet connection.	
Flows of Events:	
1. Basic Flows	
1.1 Register Account	

Step	User Actions	System Actions
1	The customer selects "Register."	The system displays the registration form with fields for first name, last name, email, phone number, driver's license number, and password.
2	The customer fills out the form and requests the system to save the details.	The system verifies that no other profile exists with the same email address and saves the profile, returning a success message and logging the customer in. In case another profile exists with the same email address, the system returns a message indicating a duplicate account exists.
Postconditions		
The customer profile is persisted in the system and the customer is authenticated.		
Business Rules		
No duplicate customer profiles. A unique profile is identified by email address. Passwords must meet minimum complexity requirements.		
1.2 Login		
Step	User Actions	System Actions
1	The customer selects "Login" and enters their email and password.	The system authenticates the credentials against the stored profile, displays the customer dashboard, including current and past rentals, on success, or an invalid-credentials message on failure. After 5 consecutive failed attempts the system temporarily locks the account.
Postconditions		
The customer is authenticated		
Business Rules		
Username and Password are required.		
1.3 Update Profile		
Step	User Actions	System Actions
1	The customer selects "Edit Profile."	The system displays an editable profile form pre-populated with the customer's details.

2	The customer updates the fields they want to change and requests the system to save the new details.	The system validates and updates the record and returns a success message or a fail message on exception.
Postconditions		
The customer profile will be updated.		
Business Rules		
The email field should not allow edit.		

UC-2: Search and Reserve Vehicle

Use Case Number: UC-2		
Name	Search and Reserve Vehicle	
Description	This use case allows a customer to search real-time vehicle availability across branches and create, modify, or cancel a reservation.	
Actors	Customer	
Preconditions		
The customer must be logged into the system		
Flows of Events:		
1. Basic Flows		
1.1 Search and Create Reservation		
Step	User Actions	System Actions
1	The customer specifies branch/location, pickup and return dates, and vehicle category, and requests to search.	The system queries the centralized database for real-time vehicle availability and displays matching vehicles with specifications, pricing, and images within 2 seconds. If no vehicle matches the criteria, the system suggests alternative dates, branches, or categories.
2	The customer selects a vehicle and requests the system to reserve it.	The system calculates the estimated total cost, reserves the vehicle, updates availability in real time, and returns a confirmation within 5 seconds. The system also sends the customer a confirmation email/SMS through the Notification Service.
Postconditions		
A reservation record is created and vehicle inventory availability is updated in real time.		
Business Rules		

The system must support at least 1,000 concurrent users. Vehicle availability must remain synchronized in real time.

1.2 Modify Reservation

Step	User Actions	System Actions
1	The customer selects an existing reservation and requests to view it.	The system displays an editable reservation form pre-populated with the reservation details.
2	The customer changes the dates, category, or branch, and requests the system to save the changes.	The system recalculates the cost, updates the reservation and inventory availability, and returns a success message or a fail message on exception.

Postconditions

The reservation will be updated.

Business Rules

There should be an existing reservation.

1.3 Cancel Reservation

Step	User Actions	System Actions
1	The customer selects an existing reservation and requests to cancel it.	The system displays a confirmation dialogue window.
2	The customer selects OK to confirm the cancellation.	The system releases the vehicle back into available inventory, applies the cancellation policy, and returns a success message or a fail message on exception.

Postconditions

The reservation will be cancelled.

Business Rules

There should be an existing reservation.

UC-3: Verify Customer Identity

Use Case Number: UC-3		
Name	Verify Customer Identity	
Description	This use case automates validation of a customer’s driver’s license and identity as part of digital check-in, replacing manual counter verification.	
Actors	Customer and Driver’s License Verification Service (Secondary/External) or Rental Agent (Secondary);	
Preconditions		
The customer must have an active, confirmed reservation.		
Flows of Events:		
1. Basic Flows		
1.1 Automated Verification		
Step	User Actions	System Actions
1	The customer uploads a photo of the front and back of their driver’s license and a selfie.	The system transmits the license data to the external Driver’s License Verification Service through a secure API.
2		The verification service validates license authenticity and expiration date and matches the selfie to the license photo. The system marks the customer’s identity as verified on success and notifies the customer.
Postconditions		
The customer’s identity verification status is recorded; check-in may proceed only if identity is verified.		
Business Rules		
Customer driver’s license should be valid		
1.2 Manual Verification		
Step	User Actions	System Actions
1	The Rental Agent reviews the uploaded documents and the physical license at the counter when automated verification is inconclusive, or the verification service is unavailable.	The system allows the Rental Agent to manually override the verification status if the documents are valid.

2	The customer changes the dates, category, or branch, and requests the system to save the changes.	The system recalculates the cost, updates the reservation and inventory availability, and returns a success message or a fail message on exception.
Postconditions		
The customer's identity verification status is recorded.		
Business Rules		
Customer driver's license should be valid		

UC-4: Complete Digital Check-In

Use Case Number: UC-4		
Name	Complete Digital Check-In	
Description	This use case enables a customer to complete rental paperwork digitally, including reviewing and signing the rental agreement, prior to vehicle pickup.	
Actors	Customer; Rental Agent (Secondary)	
Preconditions		
A confirmed reservation must exist and identity must be verified		
Flows of Events:		
1. Basic Flows		
1.1 Digital Check-In		
Step	User Actions	System Actions
1	The customer accesses check-in from the confirmation email or application.	The system includes the Verify Customer Identity and auto-generates the rental agreement using the reservation and vehicle details.
2	The customer reviews the rental terms, insurance options, and add-ons, and applies an electronic signature.	The system stores the signed digital contract, marks the reservation status as “Ready for Pickup,” and generates a QR code/pickup code. If the customer declines the contract terms, the system cancels the check-in and returns the reservation to “Pending.”
Postconditions		
A signed digital contract is stored and the customer receives a QR code/pickup code.		
Business Rules		
Customer must agree to terms and conditions.		

1.2 Agent-Assisted Check-In		
Step	User Actions	System Actions
1	The Rental Agent completes the digital check-in on behalf of a customer without app access.	The system processes the check-in.
2	The customer reviews the rental terms, insurance options, and add-ons, and applies an electronic signature.	The system stores the signed digital contract, marks the reservation status as “Ready for Pickup,” and generates a QR code/pickup code. If the customer declines the contract terms, the system cancels the check-in and returns the reservation to “Pending.”
Postconditions		
A signed digital contract is stored and a QR code/pickup code is issued.		
Business Rules		
Customer must agree to terms and conditions.		

UC-5: Process Payment

Use Case Number: UC-5		
Name	Process Payment	
Description	This use case processes rental charges, deposits, and refunds securely through an integrated third-party payment gateway and produces a digital receipt.	
Actors	Customer; Payment Service (Secondary/External); Finance Department (Secondary)	
Preconditions		
Check-in must be complete, or a vehicle is being returned and final billing is required.		
Flows of Events:		
1. Basic Flows		
1.1 Digital Check-In		
Step	User Actions	System Actions
1	The customer selects a payment method.	The system calculates total charges (rental fee, taxes, add-ons, deposit) and forwards the transaction request to the external Payment Service over an encrypted connection.

2		The Payment Service authorizes and processes the transaction. The system receives confirmation, generates a digital receipt, and emails it to the customer. If the payment is declined, the system notifies the customer and allows selection of another payment method. If the gateway times out, the system retries automatically or flags the transaction for rental agent assistance.
Postconditions		
A payment record is created and reflected in financial reports; a digital receipt is issued to the customer.		
Business Rules		
Customer should provide legitimate debit/credit card for payment		
1.2 Process Refund		
Step	User Actions	System Actions
1	A cancellation or return adjustment requires a refund.	The system, or the Finance Department, issues a refund request to the Payment Service, updates the payment record, and notifies the customer.
2	The customer reviews the rental terms, insurance options, and add-ons, and applies an electronic signature.	The system stores the signed digital contract, marks the reservation status as “Ready for Pickup,” and generates a QR code/pickup code. If the customer declines the contract terms, the system cancels the check-in and returns the reservation to “Pending.”
Postconditions		
A refund record is created and reflected in financial reports.		
Business Rules		
There should be an active successful payment.		

UC-6: Pick Up Vehicle (QR Code Scan)

Use Case Number: UC-6	
Name	Pick Up Vehicle (QR Code Scan)
Description	This use case allows a checked-in customer to retrieve their assigned vehicle quickly by scanning a QR code, minimizing counter wait time.
Actors	Customer; Rental Agent (Secondary)

Preconditions		
Digital check-in and payment must be complete.		
Flows of Events:		
1. Basic Flows		
1.1 QR Code Pickup		
Step	User Actions	System Actions
1	The customer scans the QR code/pickup code at a kiosk or through the mobile app.	The system retrieves the reservation, verifies check-in and payment status, and assigns the best available vehicle to the reservation if one has not already been assigned.
2		The system displays the assigned vehicle's parking location and access/unlock instructions. The Rental Agent or self-service kiosk releases the vehicle, and the system updates the reservation status to "Active Rental" and the vehicle status to "Rented." If the QR code is invalid or expired, the system directs the customer to the counter for manual assistance from a Rental Agent.
Postconditions		
The vehicle status is updated to "Rented" and the rental period begins.		
Business Rules		
Customer should have a valid QR Code		

UC-7: Return Vehicle and Record Inspection

Use Case Number: UC-7	
Name	Return Vehicle and Record Inspection
Description	This use case allows a customer to return a rented vehicle and enables a Rental Agent to record a digitized inspection (condition, mileage, fuel level, damage photos) at return, closing out the rental.
Actors	Customer; Rental Agent (Secondary)
Preconditions	
The vehicle must be on an active rental.	
Flows of Events:	
1. Basic Flows	

1.1 Return and Inspect		
Step	User Actions	System Actions
1	The customer returns the vehicle to a branch and notifies staff via the app or in person.	The system displays the digital inspection form pre-populated with the pre-rental inspection recorded at pickup.
2	The Rental Agent records the vehicle's condition, mileage, and fuel level, capturing photos if damage is found, and requests the system to save the inspection.	The system compares the inspection data against the pickup inspection, calculates any additional charges (late fee, damage fee, fuel charge), processes the final payment or refund adjustment, and updates the vehicle status to "Available" (or "Needs Maintenance" if damage is found), closing the reservation. If the customer disputes the reported condition, the system flags the case for Branch Manager review.
Postconditions		
The reservation is closed, vehicle inventory and status are updated in real time, and final charges are settled.		
Business Rules		
The should be an active vehicle rental.		
1.2 Self-Service Return		
Step	User Actions	System Actions
1	The customer drops off the vehicle and completes a return checklist through the app.	The system records the checklist and marks the vehicle as pending formal inspection.
2	The Rental Agent performs the formal inspection asynchronously.	The system compares the inspection data against the pickup inspection, calculates any additional charges (late fee, damage fee, fuel charge), processes the final payment or refund adjustment, and updates the vehicle status to "Available" (or "Needs Maintenance" if damage is found), closing the reservation. If the customer disputes the reported condition, the system flags the case for Branch Manager review.
Postconditions		
The reservation is closed once the formal inspection is completed.		
Business Rules		
The should be an active vehicle rental.		

UC-8: Manage Vehicle Inventory and Assignment

Use Case Number: UC-8		
Name	Manage Vehicle Inventory and Assignment	
Description	This use case enables staff to view, add, and update vehicles across the fleet, and enables the system to automatically assign the best available vehicle to a reservation based on category, branch, and availability.	
Actors	Rental Agent; Branch Manager; Fleet Manager	
Preconditions		
The vehicle must be on an active rental.		
Flows of Events:		
1. Basic Flows		
1.1 Add/Update Vehicle		
Step	User Actions	System Actions
1	Staff requests to add a new vehicle, or selects an existing vehicle to update.	The system displays a blank vehicle form (add) or an editable form pre-populated with the vehicle’s details (update).
2	Staff enters or updates the vehicle’s category, branch, and status, and requests the system to save.	The system validates and saves the vehicle record and updates availability in real time across all branches.
Postconditions		
Inventory records and vehicle status remain synchronized across branches in real time.		
Business Rules		
No duplicate vehicle information. Vehicle ID is read only		
1.2 Assign Vehicle to Reservation		
Step	User Actions	System Actions
1	A reservation reaches check-in or pickup and requires a vehicle.	The system automatically selects the best available vehicle matching the reservation’s category, branch, and availability criteria, and assigns it to the reservation.
2	The Rental Agent manually reassigns a different vehicle if the automatically assigned vehicle becomes unavailable.	The system updates the assignment and the inventory record.
Postconditions		

Inventory records and vehicle status remain synchronized across branches in real time.

Business Rules

Vehicle should be in an “Available” state.

UC-9: Schedule Vehicle Maintenance

Use Case Number: UC-9

Name	Schedule Vehicle Maintenance
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Description	This use case allows the Fleet Manager to schedule and track preventive and corrective maintenance for vehicles and to monitor maintenance status fleet-wide.
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Actors	Fleet Manager
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Preconditions

The Fleet Manager must be authenticated with appropriate role permissions.

Flows of Events:

1. Basic Flows

1.1 Schedule Maintenance

Step	User Actions	System Actions
1	The Fleet Manager reviews maintenance schedules and alerts and selects a vehicle to schedule for service.	The system marks the vehicle “Unavailable” for the maintenance window and removes it from bookable inventory.
2	The Fleet Manager updates the maintenance record and vehicle condition upon service completion.	The system returns the vehicle to “Available” status. If the vehicle requires urgent maintenance mid-rental, the system flags the reservation for Branch Manager review upon return.

Postconditions

Maintenance records are updated and vehicle availability reflects the current maintenance status.

Business Rules

Vehicle should required maintenance.

UC-10: Generate Reports and Analytics

Use Case Number: UC-10		
Name	Generate Reports and Analytics	
Description	This use case provides dashboards and reports on rental trends, fleet utilization, revenue, and branch performance, including demand forecasting.	
Actors	Branch Manager; Fleet Manager; Finance Department (Secondary)	
Preconditions		
The manager must be authenticated with appropriate role permissions.		
Flows of Events:		
1. Basic Flows		
1.1 Generate Report		
Step	User Actions	System Actions
1	The manager selects a report type (utilization, revenue, branch performance, or demand forecast) and a date range.	The system aggregates data from reservations, payments, fleet records, and vehicle-location data across all branches.
2	The manager requests to view or export the report.	The system displays the report/dashboard with visualizations and exports it as PDF/Excel if requested.
Postconditions		
A report or dashboard view is generated; no underlying data is modified.		
Business Rules		
User should be logged in.		

UC-11: Manage Users and Access

Use Case Number: UC-11	
Name	Manage Users and Access
Description	This use case allows the System Administrator to create, update, and deactivate staff user accounts and to assign role-based permissions across the system.
Actors	System Administrator

Preconditions		
The administrator must be authenticated with administrative privileges.		
Flows of Events:		
1. Basic Flows		
1.1 Create/Update User Account		
Step	User Actions	System Actions
1	The administrator selects "Manage Users" and requests to create a new account, or selects an existing account to update.	The system displays a blank account form (create) or an editable form pre-populated with the account's details (update).
2	The administrator enters or updates the account details and role (Rental Agent, Branch Manager, Fleet Manager, or Finance Department) and requests the system to save.	The system verifies there is no other account with the same email/username, saves the account, applies role-based access control, and returns a success message. In case of a duplicate email/username, the system returns a message indicating a duplicate account exists.
Postconditions		
User accounts and role permissions are updated and access control is enforced system-wide.		
Business Rules		
No duplicate users, unique username and password meets complexity requirements.		
1.2 Deactivate User Account		
Step	User Actions	System Actions
1	The administrator selects an account and requests to deactivate it.	The system displays a confirmation dialogue window.
2	The administrator selects OK to confirm.	The system deactivates the account, revokes access, and returns a success message.
Postconditions		
The account will be deactivated.		
Business Rules		
The user should exists in the system.		

5. External Interface Requirements

5.1 User Interfaces

- Responsive web application for desktop and mobile devices.
- Standard elements on every customer-facing screen: help access, account/profile menu, and current reservation status indicator.
- Screen layouts for vehicle search results must display vehicle image, category, price, and availability at a glance.
- Error messages must state the problem in plain language and the corrective action available to the user (e.g., “This email is already registered — log in instead?”).
- Detailed UI mockups and screen flows are to be documented in a separate User Interface Specification.

5.2 Hardware Interfaces

- Barcode/QR scanners
- Receipt printers
- Mobile device cameras for document upload

5.3 Software Interfaces

External System	Purpose	Interface
Payment Service	Authorizes and processes customer payments, deposits, and refunds.	Secure REST API over HTTPS; supports charge, refund, and status-query operations.
Driver’s License Verification Service	Validates license authenticity, expiration, and selfie match.	Secure REST API over HTTPS; submits document images, returns a verification result.
Notification Service	Delivers email and SMS notifications to customers.	REST API; accepts recipient, channel, and message content.
Vehicle Tracking Service	Supplies real-time vehicle location data.	REST/streaming API; returns vehicle ID and coordinates.

5.4 Communications Interfaces

- All client-to-server and server-to-external-service communications use HTTPS/TLS.
- Email notifications use standard SMTP-based delivery through the Notification Service.
- SMS notifications are delivered through the Notification Service’s carrier integration.
- API requests and responses use JSON message formatting.

6. Supplementary (Non-Functional) Requirements

The following non-functional requirements, carried forward from the Vision Document, constrain how the use cases in Section 3 must be realized.

6.1 Performance Requirements

- Vehicle search results displayed within 2 seconds.
- Reservation confirmation generated within 5 seconds.
- The system must support at least 1,000 concurrent users.
- 99.9% system availability.
- Real-time data synchronization.

6.2 Usability Requirements

- User-friendly interface requiring minimal training for self-service workflows.
- Mobile-responsive design across phones, tablets, and desktops.
- Accessibility compliance.

6.3 Security Requirements

- Multi-factor authentication for all employee accounts.
- Encryption of customer information at rest and in transit.
- Secure online payment processing.
- Compliance with data privacy regulations.
- Role-based authorization enforced system-wide.

6.4 Platform Requirements

- Web application.
- Android application.
- iOS application.
- Cloud-hosted database.
- RESTful API architecture.

7. Glossary

This glossary defines domain and business terms used throughout this document. For system acronyms and abbreviations, see the list below.

Acronym	Definition
HPCRMS	High Priority Car Rental Management System — the system described in this document.
SRS	Software Requirements Specification.
REQ	Functional Requirement identifier.
QR Code	Quick Response code.

Acronym	Definition
API	Application Programming Interface.
MFA	Multi-Factor Authentication.
RBAC	Role-Based Access Control.

Term	Definition
Actor	A person or external system that interacts with HPCRMS to achieve a goal.
Active Rental	The status of a reservation between vehicle pickup and vehicle return.
Branch	A physical rental location connected to the centralized cloud database.
Digital Check-In	The process by which a customer completes rental paperwork and signs the rental agreement electronically before pickup.
Fleet Utilization	A measure of how much of the available vehicle fleet is actively rented over a given period.
Multi-Factor Authentication (MFA)	An authentication method requiring more than one form of verification, required for all employee accounts.
Pickup Code / QR Code	A scannable code generated at check-in that a customer presents to retrieve their assigned vehicle without counter paperwork.
Rental Agreement	The digital contract generated from reservation and vehicle details that a customer reviews and signs during check-in.
Reservation	A record representing a customer's request to rent a specific category of vehicle for a given date range and branch.
Role-Based Access Control (RBAC)	A security model that restricts system functionality according to a user's assigned role.
Vehicle Assignment	The process of matching a specific vehicle to a reservation based on category, branch, and availability.
Vehicle Inspection	The recorded condition, mileage, fuel level, and (if applicable) damage photographs of a vehicle at pickup or return.
Vehicle Inventory	The complete set of vehicles owned by the company and their current status (available, rented, maintenance) across all branches.